lot or herd. These indices define herd characteristics. For large-scale decisions, the herd has to be treated as a whole: for example, if after tweaking the feed and so on, a particular herd falls below the standards, it will be culled.

Medical Records: The lifetime record of all the treatment carried out on a particular animal are held here. The dosage of commonly-used antibiotics and residues within the withdrawal period for each are also inbuilt.

Report Generation: Current or date-specified reports of milk production, breeding, bull performance, due for calving, heifers, calves, and so on can be generated category-wise.

The Chitale-owned farm at Bhilwadi with its pre-existing systems used to achieve a milk yield of 2,500 litres in a 305-day lactation cycle against the national average of 800 to 1,000 litres for a buffalo. Herdman increased the 2,500-litre figure by about 10 per cent. Also, a remarkable reduction was seen in the quantity of antibiotics that needed to be used: this was a result of Herdman's ability to predict sicknesses at very early stages. Breeding efficiency was another area of improvement.

The Next Step: Herdman^Coop

It was the creation of milk co-operatives that resulted in the present abundance of milk in India. This co-operative movement forms the backbone of the dairy industry in the country. The basic unit is the primary dairy society in a village with 500 to 1,000 animals. Several such societies federate to form the block-level or district-level federation. These federations provide milk collection as well as veterinary input services.

Herdman^Coop is a version designed to run on the Milk Union Server, with PDAs for data collection at the village levels, and touch-screen access for the village dairy farmer. While the modules are similar to those of Herdman, the critical difference is in data input and validation mechanisms, in the absence of the formal establishment of an organised farm. The zone veterinarian analyzes data for each village, considered as one herd, and works with farmers in optimising fertility and productivity to make the dairy business profitable for the farmers. In March 2004, the World Bank conducted the "India Country Development Marketplace Competition," and invited innovative and novel ideas for improving the access and quality of services in rural India. Dr Samad's proposal to implement IT-enabled veterinary services to improve animal productivity, milk quality and farmer profitability was awarded the top rank amongst 1,500 proposals received from across the country.

Supported by the NABARD (The National Bank for Agriculture and Rural Development), the co-operative version of the software has a two-year research funding for deployment, testing and validation, and is already deployed at Sangamner District Co-operative. Other than being able to improve genetics by planned breeding, it also enables the management of the health and productivity of the animals. Moreover, the data mined is of much value to pharmaceuticals developing drugs for the dairy industry, research institutes for identifying problem areas, and the government for rational planning.

Vetting For Quality

Most co-operatives have a para-vet who periodically visits the village milk collection centre. Apart from his usual duties, he is enabled with a PDA with a customised application for data collection. All permutations and combinations of entries are already fed in so he doesn't have to type but only select the activity. The para-vet is responsible for updating the (local) data he collects on the milk centre computer.

When the para-vet visits the zonal centre, he hands over a USB stick with the data he has collected to the veterinarian there, who can load it to the server linked to the Milk Union Server. For the case when these visits to the Milk Union Server are irregular, a GPRS data transmit option to the server is being developed. Data from all the Milk Union Servers will thus form a state-level animal grid. The farmers get monthly report in his language describing performance of each cow / buffalo.

A critical aspect of the system is the validity of the data submitted by the para-vet. Among the different options in data authentication is RFID-activated access to the animal data recording module in the PDA, which means an animal's file can only be accessed in the presence of its RFID tag—necessitating that the para-vet visit the animal. Whereas, as of now, the veterinarian only manages sick animals at the dispensary, the data recording system would improve services to the farmers since the protocol of services would be generated by the PDA. This will bring veterinary services in India at par with those in the developed world.

A para-vet showing off his PDA

Among the developmental issues being tackled is distance-reading capabilities for the PDA, since proximity may cause the animal to jerk or damage the PDA reader in other ways.

From the farmer's perspective, he can visit the village milk centre and access records of his animal, get information on animals available for sale and access **www.pashubazar.com**. Availability of animal milk, breeding and medical records at the time of sale would fetch the farmer better price for its cow and the purchasing farmer would remain assured that he has bought the right cow or buffalo. Touch-screen access for farmers is also under implementation—this will enable them to access data of their animals from the village milk centre computer by themselves. In developed countries, there are independent organisations for data recording, making the exercise expensive. Dr Samad, assisted by his team, has developed a system wherein the veterinary service provider captures data in the same visit.

Available in all Indian regional languages, Herdman^Coop, if successfully deployed, could change the face of dairy-farming in India right at the grassroots level: a state-level animal grid could very well bring the Indian dairy scene on par with countries such as Denmark and Canada. ■

readersletters@thinkdigit.com